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DEPARTMENT OF DEFENSE STANDARD PRACTICE

DEVELOPMENT OF SHIPBOARD INDUSTRIAL TEST PROCEDURES

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FOREWORD

1. This standard is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

2. This standard describes an engineering process for defining test requirements and developing test outlines and test procedures for use principally in the shipyard industrial environment. It includes a standardized format to be used in documenting those test outlines and test procedures. This standard will normally be invoked in its entirety when test procedures are being developed for new equipment and systems or when existing equipment and systems have undergone major modification. When technically adequate test procedures exist for the same or similar equipment and systems, the requirements of this standard may be tailored to take advantage of previous development efforts. The decision as to when and how this standard will be invoked on a given test program will be made by the appropriate program manager; however, it is neither the intent nor purpose of this standard to require needless redevelopment or reformatting of technically adequate existing test documentation.

3. Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1. SCOPE

1.1 Scope. This standard describes the engineering process for developing the content of shipboard industrial test procedures for equipment and systems.

1.2 Applicability. This standard is intended to be used by either contractor or Navy organizations in the development of industrial test procedures for shipboard equipment and systems. It applies to test procedures used during initial installation, after repair or modification, and after extended shutdown periods. The standard applies to test procedures used during ship construction, modernization, overhauls, and restricted availabilities. Unless specifically invoked by the program manager in the Overhaul Work Package, this instruction is not invoked on in-service availabilities undergoing work at maintenance shipyards. The Naval Supervisory Activity (NSA) has the responsibility to ensure and implement an adequate test program. In-Service Submarine Non-Propulsion Electronics Systems (NPES) and Hull, Mechanical & Electrical (HM&E) test procedures are exempted from this standard and are governed by the following policies:

a. NAVSEA T9SSN-W4-GYD-010, Standardized Test Program Manual (TPM) SSN Submarines

b. NAVSEA TL710-AD-MAN-010/SSBN/SSGN, 726 CL Ohio Class Submarine Test Program Manual (TPM)

The above references serve as the vehicle for the publication, dissemination, and maintenance of in-service submarine test requirements.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this standard to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE MANUALS

DOD 5220.22 - National Industrial Security Program Operating Manual

(Copies of this document are available online at http://www.dtic.mil/whs/directives.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

NAVSEA OP 1700 - Standard Fire Control Symbols Vol. 1 through 3

NAVSEA ST000-AA-IDX-010-PEETE - Electrical/Electronic Test Equipment Index

(Copies of these documents are available online at https://nll.ahf.nmci.navy.mil, may be requested by phone at 215-697-2626, or may be requested by email at nllhelpdesk@navy.mil. These publications can be located by searching the Navy Publications Index for the TMIN without the suffix.)

SECNAV INSTRUCTIONS

SECNAV M-5510.36 - Department of the Navy Information Security Program

(Copies of this document are available online at http://doni.documentservices.dla.mil/default.aspx.)

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UNITED STATES GOVERNMENT PRINTING OFFICE

United States Government Printing Office (GPO) Style Manual

(Copies of this document are available online at www.access.gpo.gov.)

2.3 Non-Government publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ISO 14617 series - Graphic Symbols for Diagrams

ANSI/IEEE STD 315 - Graphic Symbols for Electrical and Electronic Diagrams

(Copies of these documents are available online at http://webstore.ansi.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Characteristic. Any quality, property, capacity, or trait of an equipment or system.

3.2 Combat system. A combination of equipment and systems that operate in concert to accomplish the functions of detecting, tracking, identifying, processing, evaluating, and controlling the engagement of hostile threats either actively or passively.

3.3 Contract data requirements list (CDRL) and data item description (DID). A CDRL is a contract form which provides, in one place in the contract, a list of data items required to be delivered under the contract. DD Form 1423 is used for this purpose. A DID is a contract form which defines the format and content of a single deliverable data product. DD Form 1664 is used for this purpose.

3.4 Equipment. A single unit of hardware or a closely related group of hardware units that is essentially irreducible for purposes of testing.

3.5 Function. Any action or operation that an equipment or system is intended to perform that can be demonstrated through testing.

3.6 Inspection. A qualitative examination of the physical aspects of production, construction, and installation; e.g., workmanship, completeness of work, material conditions, physical location.

3.7 Interface. A connection between two equipment items or systems.

3.8 Master test plan (MTP). The MTP, if contractually required, is a collection of test outlines that defines the test program for a given system or program. It is intended as a reference document that an agency and test team can consult to coordinate and plan test activities.

3.9 Shipboard industrial test. A test conducted aboard ship to verify acceptable operation following installation, reinstallation, overhaul, repair, or periods of inactivity.

3.10 Ship system. A combination of equipment that operates in concert to accomplish all of the active and passive functions associated with propulsion, electrical, hull, or auxiliary features of a ship.

3.11 System. A combination of two or more equipment items that operate in concert to accomplish an operational function or set of functions.

3.12 Test. A predetermined set of conditions, operations, and measurements that enable quantitative assessment of equipment and system operability and readiness to support mission objectives. Tests are documented by test procedures and performed as part of the industrial test program.

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3.13 Test change proposal (TCP). A form used for the proposal and approval of test documentation changes prior to shipboard testing (see figure 1).

3.14 Test elements. Functions, characteristics, or interfaces that may be verified or demonstrated through testing.

3.15 Test outline. A summary description of a single test procedure that provides advanced planning information. For test documentation prepared in accordance with this standard, the test outline is the COVER SHEET, REVISION RECORD, and GENERAL INFORMATION contained at the front of each test procedure. Test outlines include information required to plan for and conduct the test; e.g., test objective, estimated testing time, references, test equipment requirements, computer programs required, prerequisite testing requirements, equipment involved in the test, method statement, and station assignments.

3.16 Test problem report (TPR). A form used to document test procedure, equipment, or computer program problems discovered during test conduct (see figure 1).

3.17 Test procedure. A detailed step-by-step description of the operations to be performed during the conduct of a specific test that allows for demonstration of acceptable installation and operation of shipboard elements and sub-elements. The total set of test procedures for a specific equipment shall demonstrate acceptable operation of the equipment from initial installation through integration of the equipment into the overall ship system. Test procedures include information and forms required to plan for and conduct the test; e.g., all of the test outline elements plus additional instructions, safety instructions, initial conditions and setup instructions, detailed steps to accomplish the testing, shutdown and securing instructions, test data recording sheets, test equipment used, and comments.

3.18 Validation. Verifying the feasibility, safety, accuracy, and efficiency of a test procedure through conduct in the environment for which the test was written. Validation is one method of verification.

3.19 Verification. A process by which the feasibility, safety, accuracy, and efficiency of a test procedure is evaluated through performance of the test. The test procedure may be performed aboard ship, in a land-based test site or industrial plant, or at some other appropriate facility.

4. GENERAL REQUIREMENTS

4.1 Roles and responsibilities.

4.1.1 Ship program managers (SPMs). SPMs for new construction shipbuilding programs shall ensure this standard is invoked in shipbuilding contracts and in Ship Project Directives (or equivalent) that establish a Participating Acquisition Resource Manager's (PARM) responsibilities for Government Furnished Equipment (GFE) and Government Furnished Information (GFI) in support of a new construction ship. Test procedures in accordance with this standard shall be mandatory GFI for all GFE provided for new construction ships and mandatory Contractor Furnished Information (CFI) for all Contractor Furnished Material (CFM).

4.1.2 Participating acquisition resource managers (PARMS). Each PARM that fields systems for shipboard installation shall designate a Test Development Manager (TDM) and a Test Development Agent (TDA).

4.1.3 Test development manager (TDM). The PARM shall designate a TDM who shall be responsible for the technical adequacy of all test procedures provided for use in accordance with this standard. The TDM should be the PARM's Test and Evaluation (T&E) Manager, or the T&E Manager's designated representative on the PARM staff. The TDM shall provide technical guidance, information, and management oversight of the TDA. The TDM provides a review process for Test Outlines and Test Procedures produced in accordance with this standard.

4.1.4 Test development agent (TDA). The PARM shall designate a TDA who shall report to the TDM, and is responsible for the test development engineering process and produces test procedures in accordance with this standard. The TDA will normally be an In Service Engineering Agent (ISEA) staff member with detailed technical knowledge of the system and experience in T&E.

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4.2 Security classification. The security classification of material used for development of test documentation shall be as specified (see 6.2). The use of classified test documentation shall be avoided whenever possible. If the use of classified material cannot be avoided, whenever possible, the test procedure must be written in such a manner that classified sections can be kept separate and provided as classified "addendums" to the unclassified test procedure (see 3.3). The developing organization shall have initial responsibility for determining the security classification of test documentation. Approval of this classification shall be the responsibility of the acquisition command or agency.

4.3 Test environment. Each test procedure developed in accordance with this standard shall be designed for use in the shipyard industrial test environment aboard a ship. Consideration shall be given to safety, power, ventilation, cooling water, shipyard personnel, and other factors that may distinguish the shipboard environment from the factory or shop environment. The final test procedures used aboard ship shall not contain information unique to off-ship test facilities.

4.4 Text style. Test documentation shall be prepared in accordance with the style specified in the following paragraphs.

4.4.1 Wording of text. Procedural steps shall be presented in language which is free of vague and ambiguous terms and which uses simple words and phrases to convey the intended meaning. The following guidelines shall be applicable when developing text for test documents:

a. Omit theory

b. Avoid phraseology requiring specialized knowledge

c. Avoid superfluous and ambiguous words and phrases

d. Write sentences as short and concise as possible

e. Use consistent phraseology throughout the text for common actions or events

f. Omit unnecessary articles and pronouns

g. ACCEPTABLE: Observe indicator blinks

h. NOT ACCEPTABLE: Observe that the indicator blinks

4.4.2 Grammatical person and mode. The second person imperative shall be used for test procedure operations. For example, "Remove rammer tank fluid filler cap." Description and observation statements should be written in present tense with the assumption that everything is operating correctly. For example, "Observe indicator blinks" and not "Indicator should blink."

4.4.3 Level of writing. The content of the test procedures shall be written at a level that is suitable for a test conductor who is familiar with similar types of equipment and has shipboard industrial testing experience.

4.4.4 Nomenclature. Consistency in the use of words and phrases is essential in order to achieve clarity. The following requirements shall apply:

a. Be consistent in the use of words. For example, a part identified initially as a cover shall not be identified thereafter as a plate.

b. Terms used shall agree with the same name on related engineering and design drawings and on identification plates. Put numerical designators first and descriptive nomenclature following (for example, AN/SPQ-9B Radar Set). If there is no numerical designator, then use terms such as Converter, Angle Data or Turbine Generator, and Ship Service. This allows the least significant descriptive words to be dropped as necessary to accommodate computer formats.

c. When two or more items have the same nomenclature, modifiers shall be added to define the function or location of each item. For example, AN/SPA-25 (CIC) and AN/SPA-25 (Bridge); Davit – 5000 lbs, Port and Davit – 5000 lbs, Starboard.

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d. The first time an equipment item is addressed it shall be fully identified. For example, AN/SPS-73(V) 12 Radar System, or SSDS MK 2 MOD 4B Combat Direction System introduces the equipment by its full official nomenclature. The Army/Navy designation or MK and MOD designations may be omitted thereafter or an abbreviation may be used, such as SPS-73 or SSDS MK 2. Abbreviations or acronyms shall be spelled out on first use; for example, Ship Self Defense System (SSDS).

e. Where more than one MOD of equipment is described, each MOD shall be indicated; for example, SSDS MK 2 MOD 3 and 3A.

4.4.5 Capitalization and punctuation. The Government Printing Office (GPO) Style Manual shall be used as a guide for capitalization and punctuation.

4.4.6 Spelling. Spelling should be in accordance with general Navy and standard dictionary usage.

4.4.7 Abbreviations. The use of abbreviations should be avoided where practical. When used, abbreviations should reflect general Navy usage and may be used only with permission of the contracting organization. If abbreviations are used, a list of abbreviations should be provided as supplemental information in paragraph 12 of the GENERAL INFORMATION. See figure B-9 for an example.

4.4.8 Signs and symbols. The use of symbols shall be avoided whenever practical. When used, signs and symbols shall be as listed in the GPO Style Manual. The following additional rules shall be observed in the use of symbols:

a. Fire control symbols shall be in accordance with NAVSEA OP 1700.

b. Chemical symbols shall not be used unless they form part of the nomenclature. In lieu of symbols, use the chemical names such as carbon dioxide and hydrogen peroxide.

c. Graphic symbols shall be in accordance with ANSI/ISO 14617 series and ANSI/IEEE STD 315 for electrical and electronic diagrams.

4.4.9 Type style. Unless otherwise specified (see 6.2), type font shall be sans serif generic 11 point such as Arial, Calibri, or similar. Consideration should be given to consistency of type style, applicability for automatic input to a word processing system, and ease of reading in dimly lighted shipboard spaces.

4.4.10 Spacing requirements. Test procedures shall be single-spaced with an extra line space between paragraphs, subparagraphs, testing steps, warnings, cautions, or notes.

4.4.11 Page numbering. Pages shall be numbered consecutively beginning with the cover sheet. The first page number shall also indicate the total number of pages in the document as follows: Page 1 of _.

4.4.12 Electronic media-developed test procedures. Test procedures can be developed in a variety of electronic media formats and should contain the entire original, revised, or changed test procedure. The delivered file should be converted to a standardized document exchange format such as Adobe® Acrobat® Portable Document Format (PDF) so that the end user(s) can use the test procedure without the requirements of having the same program(s) used to develop the procedure. This also allows the file to be printed without rearranging or reformatting pages.

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5. DETAILED REQUIREMENTS

5.1 Test stages. Testing of shipboard equipment and systems is conducted in stages with each progressive stage building upon the results of testing at prior stages. Thus, overall system integrity is based upon the results of all required stages of testing. The seven discrete stages of industrial testing are defined in Appendix A with examples provided for stages two through seven. This standard is applicable to tests in stages two through seven.

5.1.1 Grooming. General equipment grooming, alignment, and adjustments are considered to be part of the industrial work and are not normally included in the test procedure.

5.2 Engineering process. The engineering process specified in this section provides a structured method for the development of test procedures that will demonstrate the initial acceptable performance of equipment and systems in ships from test stages two through seven. This process is depicted in block form on figure 2 and consists of the following steps:

a. Equipment/system definition
b. Test element identification
c. Test element selection
d. Test stage determination
e. Redundancy analysis
f. Test element grouping
g. Test outline preparation
h. Test outline review and approval
i. Test procedure development
j. Test number/test title assignment
k. Test procedure review and approval
l. Test procedure verification
m. Final test procedure development

5.2.1 Equipment and system definition. The initial step in the development of test procedures is to identify the hardware to be tested. This hardware is then classified as an equipment to be tested at stage three or a system to be tested at stages four and five. Classifying the hardware as a system does not imply that stage three tests will not also be required, and classification as an equipment does not imply that stages four and five tests will not be required. The equipment or system classification is based on previous test experience and intimate knowledge of the hardware and the ship environment. The following factors shall be considered in making this determination:

a. Hardware configuration and packaging
b. Functional compatibility of the hardware packages
c. Complexity of the individual hardware packages
d. Complexity of the larger system of which this hardware is a part, or the complexity of the components that comprise this hardware packaging
e. Shipboard test conditions that favor testing at one stage over another

Hardware classified as equipment or system shall be defined in a breakdown diagram. When specified (see 6.2), breakdown diagrams shall be submitted.

5.2.1.1 Breakdown diagram content. The equipment or system to be tested shall be represented by one or more functional blocks enclosed in heavy or double lines. Each block shall represent an independent functional unit. External equipment or systems that interface with the equipment or system to be tested shall also be shown. External and internal interfaces shall also be shown on breakdown diagrams.

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5.2.1.2 Interfaces. Equipment and systems are connected by interfaces. Interfaces can be electrical, optical, liquid, gas, or mechanical. Each type of interface shall be indicated by a separate line. Internal interfaces are defined as those interfaces that are within the system to be documented. External interfaces are defined as those that functionally tie the equipment or system to be tested with other equipment or systems. Services with organic ship systems such as 400-Hertz power, 28 VDC, and cooling water are external interfaces. Services may be designated on the breakdown diagram by a single line with a descriptive title. External interfaces shall be designated by numbers and internal interfaces shall be designated by letters.

5.2.2 Test element identification. After the equipment or system has been defined in a breakdown diagram, test elements shall be identified. A test element is any property of an equipment or system that can be evaluated through testing. Test elements are identified by performing a functional analysis, characteristic analysis, and interface analysis. Instructions for performing these analyses are provided in 5.2.2.1, 5.2.2.2, and 5.2.2.3, respectively. Each analysis will result in a list of test elements. Each list shall then be screened to select those test elements that are appropriate for industrial testing in accordance with the instruction provided in 5.2.3.

5.2.2.1 Functional analysis. An analysis shall be accomplished to identify each function performed by the equipment or system to be tested. The functions of an equipment or system are the actions or operations it is intended to perform. Each function so identified represents a test element to be considered for inclusion in the test outlines and test procedures to be developed. In performing this analysis, the following approach shall be taken to ensure that the list of test elements generated will be comprehensive and accurate:

a. Identify the major purpose of the equipment or system to be tested. If the purpose is complex, it may be divided into two or more discrete functions.

b. List all other significant actions or services performed by this unit.

c. Examine each block on the breakdown diagram to ensure that all actions or services performed by the equipment that comprise the system under test are reflected in the list of functions.

d. Use the information developed in steps a through c above to generate a list of test elements.

Figure 3 is a sample functional analysis.

5.2.2.2 Characteristics analysis. An analysis shall be performed on the equipment or system to be tested and its parts (represented by blocks on the breakdown diagram) to identify those characteristics that significantly impact the manner and degree to which this equipment or system performs its intended function. The procedure for conducting this analysis is to analyze each breakdown diagram block comprising the equipment or system to be tested. This analysis may involve only one block for simple equipment or several blocks for complex systems. In this analysis, each salient characteristic is screened in accordance with the following criteria:

a. Does this characteristic determine how well the intended function is performed?

b. Does this characteristic distinguish a properly operating equipment or system from one that is not?

c. Does this characteristic require adjustment or alignment to ensure proper equipment or system functioning?

A positive response to any of the above questions is indicative of a characteristic that should be included in the list of test elements. If a system is being analyzed, the system shall then be considered as a whole. Any characteristics of the system that have not been previously identified through analysis of its component parts should be analyzed in accordance with steps a through c above. The separate lists of characteristics generated when a system is analyzed in this manner shall be combined to form one consolidated list of characteristics. Each of these characteristics represents a test element that shall be considered for inclusion in the test outlines and test procedures. Figure 4 is a sample characteristic analysis.

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5.2.2.3 Interface analysis. When breakdown diagrams are prepared, the mechanical, electrical, liquid, optical, and gas interfaces are each represented by a single line. Each single line may be composed of several signals and connections between equipment. When specified (see 6.2), interface analysis records shall be submitted. The first step in interface analysis shall be to list on an interface analysis form, all of the electrical or optical signals and liquid, gas, or mechanical connections that comprise each interface. Each item listed on the interface analysis form shall be identified by a brief descriptive title that focuses on its salient characteristics. Individual items shall then be analyzed to determine if they are appropriate for testing. Guidance to make this determination shall be based in the following questions:

a. Must this interface be tested in order to demonstrate satisfactory equipment performance?

b. Does this interface require checkout prior to equipment operation?

c. Would testing of this interface provide information concerning a potentially unsafe or hazardous condition of the equipment?

d. Is measurement of this interface a necessary prerequisite to further testing?

Responses to these questions shall be indicated on the interface analysis form. A positive response to any of these questions is an indication that the interface should be considered for testing. The result of interface analysis is a list of recommended interface test elements.

5.2.3 Test element selection. The lists of test elements resulting from functional, characteristics, and interface analyses shall be screened to determine which elements are appropriate for industrial testing. The objective of industrial testing is to demonstrate the ability of shipboard equipment and systems to perform their intended function. Compliance with design specifications shall be assumed to have been demonstrated in previous testing. Consequently, only those elements that meet one or more of the following criteria shall be selected for inclusion in the test outlines and test procedures to be developed.

a. Elements that are essential to equipment performance and operation.

b. Elements whose performance may be seriously degraded by shipping, installation, or shipboard environment.

c. Elements whose performance is affected by post-installation adjustments.

d. Elements whose performance is essential to overall system performance or to operation of other equipment.

e. Elements that provide indication of safe operation.

f. External interface elements that must be verified prior to operation of the equipment under test.

The results of this analysis shall be a composite list of test elements selected for industrial testing.

5.2.4 Test stage determination. When preparing test procedures and test outlines for a test stage, the TDA shall decide which test elements are appropriate for evaluation at that stage and which elements should be evaluated at higher or lower stages.

5.2.4.1 Test stage guidelines. As a general rule, any function, characteristic, or interface required to adequately demonstrate equipment performance shall be tested at stage three, system performance at stage four, and intersystem performance at stage five. Test elements that require special simulation facilities or significant resources external to the immediate testing location shall be tested at stage six. Test elements which require testing at sea shall be evaluated at stage seven. Test elements evaluated in lower level tests that need to be re-evaluated during trials should be designated for testing at both stages. Although seven stages of testing are defined, it is not intended to require testing at each stage for each equipment or system. For example, stand-alone equipment may be tested at stage three only and other systems may not lend themselves to equipment level testing.

5.2.4.2 Test stage assignment. Using test stage definitions found in Appendix A and the test stage guidelines of 5.2.4.1, the test elements shall be separated into two lists: (1) those elements which are appropriate for testing at the stage for which test procedures are being prepared, and (2) those elements which are appropriate for testing at another stage for which the TDA is not responsible. When specified (see 6.2), those test elements that are candidates for testing at another stage for which the developer is not responsible shall be listed with their recommended test stage and submitted.

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5.2.5 Redundancy analysis. Prior to preparation of test outlines, the list of test elements shall be reviewed to eliminate redundant testing within the stage for which documentation is being prepared. Each characteristic designated for testing at a given stage shall be analyzed to determine if the adequacy of that characteristic would be implicitly demonstrated by successful performance of a function designated for testing at that same stage. Characteristics meeting these criteria may be eliminated from further consideration as test elements. Interface test elements should be compared with the list of functions and characteristics designated for testing at the same stage. Interface test elements whose adequacy would be implicitly demonstrated by successful performance of a previously identified function or characteristic may be eliminated.

Elimination of all testing redundancies is not mandatory. On occasion, the added confidence gained through redundant testing outweighs the savings in reduced test time. This tradeoff requires the exercise of good judgment by the TDAs. When the redundancy analysis has been completed, the result shall be elimination of redundant test elements, wherever practical.

5.2.6 Test element grouping. The test elements that remain after completion of redundancy analysis shall be optimized for efficient test conduct by combining elements into test groups. These test element groups shall be converted to test outlines and subsequently to test procedures on a one-to-one basis; therefore, considerable care shall be exercised to group test elements in a manner that will produce maximum shipboard testing efficiency. The TDA shall consider similarities in the following areas as a guide to grouping test elements.

a. Test approach.
b. Inputs from other systems or services such as steam, cooling water, 60-Hertz power or 400-Hertz power.
c. Computer programs.
d. Special support equipment and resources.
e. Other practical considerations identifiable at this point.

Examples of groupings include (1) all elements associated with gun loading; (2) all elements associated with initial checkout of a countermeasures receiver; (3) all elements associated with launcher hydraulics; and (4) all elements associated with initial start up of a diesel or gas turbine. This redundancy analysis will result in one or more groups of test elements which shall be the basis for preparation of test outlines. When specified (see 6.2), a list of non-redundant test elements, which is the basis for proposed test outlines, shall be submitted prior to their preparation.

5.2.7 Test outline preparation. A test outline shall be prepared as prescribed by Appendix B for each test element group identified in 5.2.6. The test outline may be used to:

a. Allow the test contracting agent to review the test development approach prior to test procedure completion
b. Provide the industrial activity preliminary information for scheduling and manpower planning
c. Supply a description of test procedure content for all levels of management
d. Provide preliminary bidding information

The information provided in the test outline shall be based on the best information available at this point in the development cycle and shall be supplemented by sound engineering judgment.

5.2.8 Test outline submittal. When specified (see 6.2), test outlines (see 5.2.7) and the list of related test elements (see 5.2.4.2) shall be submitted.

5.2.8.1 Master test plan submittal. When specified (see 6.2), test outlines (see 5.2.7) shall be combined into a CDRL document and submitted in accordance with contract specifications.

5.2.9 Test procedure development. When specified (see 6.2), after submittal of a test outline, a stand-alone test procedure shall be developed from the test outline. The Test Number/Test Title will be part of the test procedure development process as described in Appendix A. The format and content of each test procedure shall be as prescribed in Appendix B.

Because the contents of all other test procedure sections depend to some degree on the testing steps, test procedure preparation shall begin with the development of the Testing Steps section.

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5.2.9.1 Testing steps. The Testing Steps section of each test procedure shall provide detailed instructions for test implementation. Each testing step in the test procedure shall be defined and sequenced to ensure that the test procedure can be efficiently performed. The METHOD section of the GENERAL INFORMATION provides the starting point for testing steps development; each test step represents an expansion and detailed explanation of the general approach presented in the METHOD section. In many cases these steps will have been previously developed in other similar test procedures. Testing steps from similar procedures should be examined to determine if they could be modified to fulfill the current requirements. Some potential sources of testing steps are factory acceptance test procedures, maintenance requirement cards, technical repair standards, technical manuals, specifications, standards, and test document repositories. The procedure as written shall be self-contained and not refer to steps, paragraphs, or sections of technical manuals or other publications. When testing steps are being developed or modified and when the optimum sequence is being determined, the following factors should be considered:

a. Economical use of test personnel.

b. Use of standard shipboard test equipment as defined in NAVSEA ST000-AA-IDX-010-PEETE.

c. Economical use of support equipment.

d. Minimizing the requirements for external support.

e. Steps must be practical to perform in a shipboard environment.

f. Do not include steps that require adjustment of the equipment under test.

Testing steps shall be documented as specified by Appendix B for format and content.

5.2.9.2 Test data recording sheet. The TEST DATA RECORDING sheet provides a place for recording each indication required in the testing steps and provides a clear statement of expected results including applicable tolerances. After the testing steps have been developed, the most critical aspect of test procedure development is to specify the expected results. Data to be used in the establishment of expected results can be found in operational requirements documents, specifications, factory acceptance test criteria, maintenance requirement cards, technical repair standards, and technical manuals; however, the tolerances found in these documents may require modification prior to use in industrial test procedures. The purpose of industrial testing is not to determine if the system or equipment is performing to specification (factory acceptance tests); rather, the purpose is to verify that the system or equipment, as installed, will meet performance requirements for that ship. Therefore, the tolerances used for shipboard industrial testing may be different from those found in specifications or in factory acceptance tests. Performance limits for shipboard industrial testing shall be established such that performance outside these limits would be indicative of one or more of the following factors:

a. Serious degradation in the ability of the system or equipment to perform its intended function.

b. Abnormal or marginal operating condition that may indicate or result in equipment failure or low post-repair reliability or functionality.

c. Inability of the system or equipment to interface properly with other equipment or systems.

These performance limits become the expected results and appropriate tolerances for the EXPECTED RESULTS column of the TEST DATA RECORDING sheet. The format and content for the TEST DATA RECORDING sheet shall be as specified in Appendix B.

5.2.9.3 Test conditions and equipment. After the testing steps have been specified and arranged in the appropriate sequence, the INITIAL CONDITIONS AND SETUP, SHUTDOWN AND SECURING, and TEST EQUIPMENT USED sheets shall be prepared. Format and content guidance for preparation of these paragraphs is included in Appendix B.

5.2.9.4 General instructions. The GENERAL INFORMATION section of the test procedure provides the general instructions relating to the TESTING STEPS section. The general instructions provided by the GENERAL INFORMATION section shall be reviewed and updated prior to preparation of the test procedure to ensure that these instructions are consistent with the testing steps developed in 5.2.9.1. Safety instructions shall be added to the test procedure in accordance with the format and content requirements provided in Appendix B.

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5.2.10 Initial test procedure submittal. When specified (see 6.2), existing test procedures shall be submitted for approval for reapplication in lieu of the preparation of new test procedures. In the absence of technically adequate existing test procedures or as specified (see 6.2), preliminary new test procedures shall be submitted prior to verification.

5.2.11 Test procedure verification. All test procedures shall be verified. The purpose of verification is to:

a. Ensure that the equipment or system can be safely tested

b. Affirm that the testing steps can be performed as written and in the specified sequence

c. Determine if the testing steps represent the most efficient and practical method for accomplishing the desired results

d. Determine if extraneous influences exist that may prevent the accomplishment of the intended results

e. Verify the expected results presented in the TEST DATA RECORDING sheets

f. Verify SAFETY INSTRUCTIONS, TEST EQUIPMENT USED, and other pertinent sections of the test procedure.

5.2.11.1 Verification process. Verification requires that the testing steps be performed in the indicated sequence on an Element/Sub-element that is as close to the shipboard configuration as possible to accomplish the purposes listed above. Problems shall be corrected and potential improvements shall be incorporated in the final document. Two approaches to verification are:

a. Verification of test procedures during first time use in a shipboard test program (validation).

b. Verification of test procedures prior to use by performing the test in an industrial plant, at a land-based test site, or aboard a ship.

The type of verification required for each test procedure shall be as specified (see 6.2). If verification is accomplished through validation, the discrepancies and problem will be identified and corrected through the established change control process. If verification is accomplished prior to use, a plain paper or marked-up test procedure shall be completed and the procedures of 5.2.11.2 through 5.2.12.2 shall be followed.

5.2.11.2 Verification analysis record submittal. Verification Analysis provides a means of finding and documenting problems and potential improvements in draft test procedures. A Verification Analysis Record shall be created for each draft test procedure. Instructions for completion of this record are provided in the following paragraphs. When specified (see 6.2), Verification Analysis Records shall be submitted in accordance with contract specifications.

5.2.11.2.1 Time. Beginning and ending times shall be recorded and elapsed time calculated for comparison with the estimated testing time given in the test procedure. The elapsed time shall include setup and shutdown procedures. If testing is interrupted prior to completion, multiple start and stop times shall be recorded to permit calculation of total elapsed time required for the test.

5.2.11.2.2 Test steps. As they are performed, each testing step shall be examined for safety, necessity, practicality, and proper sequence. Guidelines for this examination are as follows:

a. Safety. Can this step be performed without potential hazard to personnel or equipment? Is the step sufficiently trouble-free so that inexperienced personnel would not require any special warnings or cautions?

b. Necessity. Consider why this step is being performed. Is it essential to proper performance of this test?

c. Practicality. Can this test be performed in the manner described by the test procedure? Is this the most practical way to perform the step?

d. Sequence. Consider why this step is being performed at this point in the procedure. Is this the most appropriate place within the procedure to perform this step?

e. Ambiguity. Can the instruction be misinterpreted? Is it subject to more than one interpretation?

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Any problems noted shall be documented by listing the testing step involved and providing a brief description of the problem. If the solution to the problem is evident, it shall be entered as a RECOMMENDED CHANGE. If testing steps do not contain a specific problem but could be improved, then enter the appropriate information as a RECOMMENDED CHANGE.

5.2.11.2.3 Test data discrepancies. When a test procedure is being verified and the actual test results do not agree with the expected results, both results shall be recorded in the Verification Analysis record. Differences in test environment, equipment status, or other test conditions that may have contributed to this discrepancy should also be documented.

5.2.11.2.4 Support. If actual requirements for test equipment, computer programs, or personnel differ from those listed in the test procedure, the differences will be recorded in the Verification Analysis record. A brief description of the discrepancy should be provided.

5.2.12 Final test procedure development. Final test procedures shall be developed by incorporation of improvements resulting from test procedure verification. The finalization process requires three steps: (1) analysis of verification records, (2) revision of test procedures, and (3) re-verification of updated test procedures.

5.2.12.1 Verification record analysis. Verification records for every test procedure shall be reviewed by the test documentation developer to determine the modifications required to optimize that procedure. Elapsed time shall be checked against the estimated time listed in the test procedure and deviations of greater than 10 percent shall be considered modification. Testing steps that are unsafe or present risk to equipment or personnel shall be changed or have the required warnings and cautions added. As necessary, impractical steps shall be altered, unnecessary steps eliminated, and improper testing sequences reordered to make the procedures practical and efficient. Test data discrepancies shall be analyzed to determine if the discrepancy was due to test conditions or test procedure error. Omissions in the areas of test equipment, computer programs, or personnel shall be corrected and unnecessary requirements shall be eliminated. The TDA or equipment ISEA must resolve any such discrepancies to ensure the expected results or support requirements accurately reflect the technical requirements for equipment installed in shipboard spaces.

5.2.12.2 Re-verification. In cases where the verification resulted in a major revision of a test procedure, re-verification may be required. The decision on whether or not to re-verify shall be made by the procuring organization and will be based on the extent to which revised testing steps required new or different methods. Unless these new methods or steps are sufficiently complex to require verification through actual performance, re-verification shall not be required.

5.3 Final documentation submittal. When specified (see 6.2), submittal of final test procedures and test outlines shall consist of a digital copy prepared in Microsoft Word or as an Adobe® Acrobat® Portable Document Format (PDF).

5.4 Quality assurance. Document quality assurance shall be the responsibility of the TDA and shall include the inspection and review requirements specified in the following paragraphs. The contracting organization will participate to the extent provided (see 6.2).

5.4.1 Inspection of documentation. Each test outline and test procedure submitted by terms of the contracting document as deliverable items shall be inspected for compliance with the requirements of this standard. Unless otherwise specified (see 6.2), the TDA shall be responsible for the performance of this inspection.

5.4.2 Quality assurance in-process review. The contracting organization or designated agent will perform in-process reviews. The test document developer shall provide support and material for these reviews as required.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This standard is intended to be used by either contractor or Navy organizations in the development of industrial test procedures for shipboard equipment and systems.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this standard.

b. The security classification of material used for development of test documentation (see 4.2).

c. Type font other than that specified (see 4.4.9).

d. When submittal of breakdown diagrams is required (see 5.2.1).

e. When submittal of interface analysis records is required (see 5.2.2.3).

f. When test element listing is required (see 5.2.4.2).

g. When submittal of list of non-redundant test elements is required (see 5.2.6).

h. When a list of test elements is required (see 5.2.8).

i. When test outlines are required (see 5.2.8.1).

j. When a stand-alone test procedure developed from the test outline is required (see 5.2.9).

k. When submittal of existing test procedures is required (see 5.2.10).

l. When submittal of preliminary new test procedures is required (see 5.2.10).

m. When verification for each test procedure is required (see 5.2.11.1).

n. When submittal of Verification Analysis Records is required (see 5.2.11.2).

o. When submittal of final test procedures and test outlines are required (see 5.3).

p. The extent of the contracting organization participation in document quality assurance (see 5.4).

q. Responsibility for the performance of inspection of each test outline and test procedure, if other than the TDA (see 5.4.1).

6.3 Subject term (key word) listing.

Test, format

Test, number

Test, title

6.4 Data Item Descriptions (DIDs). This standard has been assigned an Acquisition Management Systems Control (AMSC) number authorizing it as the source document for the following DIDs. When it is necessary to obtain the data, the applicable DIDs must be listed on the Contract Data Requirements List (DD Form 1423).

DID Number	DID Title
DI-QCIC-80202A	Breakdown Diagram
DI-QCIC-80203A	Interface Analysis
DI-QCIC-80204A	Test Element List
DI-QCIC-80205A	Ship Test Outline
DI-QCIC-80206A	Shipboard Industrial Test Procedure
DI-QCIC-80207A	Verification Analysis Record

The following is a list of associated test DIDs:

DID Number	DID Title
DI-MGMT-81643	Ship Trial Agenda

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DI-MISC-81615	Craft Trial Report
DI-MISC-81620	Craft Test Status Report
DI-MISC-81621	Test Change Proposal
DI-MISC-81622	Test Problem Report (TPR)
DI-MISC-81624	Notification of Test/Trials
DI-MISC-81630	Craft Acceptance Test Index
DI-QCIC-80553	Acceptance Test Plan

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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TEST PROBLEM REPORT/TEST CHANGE PROPOSAL

SUPPLEMENTAL LETTER SIZE SHEETS SHALL BE UTILIZED, IF NECESSARY

HULL NUMBER	☐ TEST CHANGE PROPOSAL ☐ TEST PROBLEM REPORT	FILE NUMBER
FROM (ACTIVITY, INCLUDE ZIP CODE)		SERIAL NUMBER
		DATE
TO (ACTIVITY, INCLUDE ZIP CODE)		LOG #

TEST PERIOD	TEST DOCUMENTATION		DATE DISCOVERED
☐ SCN	TEST NUMBER	REVISION	INDIVIDUAL CONTACTED
☐ LIFE CYCLE	TEST TITLE		DATE CONTACTED

LOCAL ACTION TAKEN	PROBLEM DESCRIPTION/REASON FOR CHANGE ☐ SEE ATTACHED SHEET	
☐ AUTHORIZED INTERIM SOLUTION		
☐ CLOSED OUT	NAME (TD) (PRINTED) DATE INITIAL	
☐ NO SOLUTION AVAILABLE	LOCAL SOLUTION (STATE SOLUTION OR PATH BEING PURSUED TO RESOLVE PROBLEM) ☐ SEE ATTACHED SHEET	
	NAME (ENG) (PRINTED) DATE INITIAL	

TDD/TDA ACTION TAKEN	CLOSE OUT	GATEWAY/TRANSMITTAL DATE
☐ CONCUR	☐ SATISFACTORY.. REMOVE FROM PROBLEM STATUS SUMMARY	
☐ DO NOT CONCUR	☐ FTSN IMPACT	
☐ CONCUR WITH EXCEPTIONS	AGENCY THAT PROVIDED SOLUTION	GATEWAY/TRANSMITTAL #
☐ SEE ATTACHED SHEET		
☐ HULL APPLICABILITY -	TDA	DATE
PREPARED BY:	DATE	TDD DATE
LCSTDD	DATE	CLOSED BY DATE
EMAIL	PHONE	

TPR/TCP FORM 08/04 Page 1 of _____

FIGURE 1. Test problem report/test change proposal sample form.

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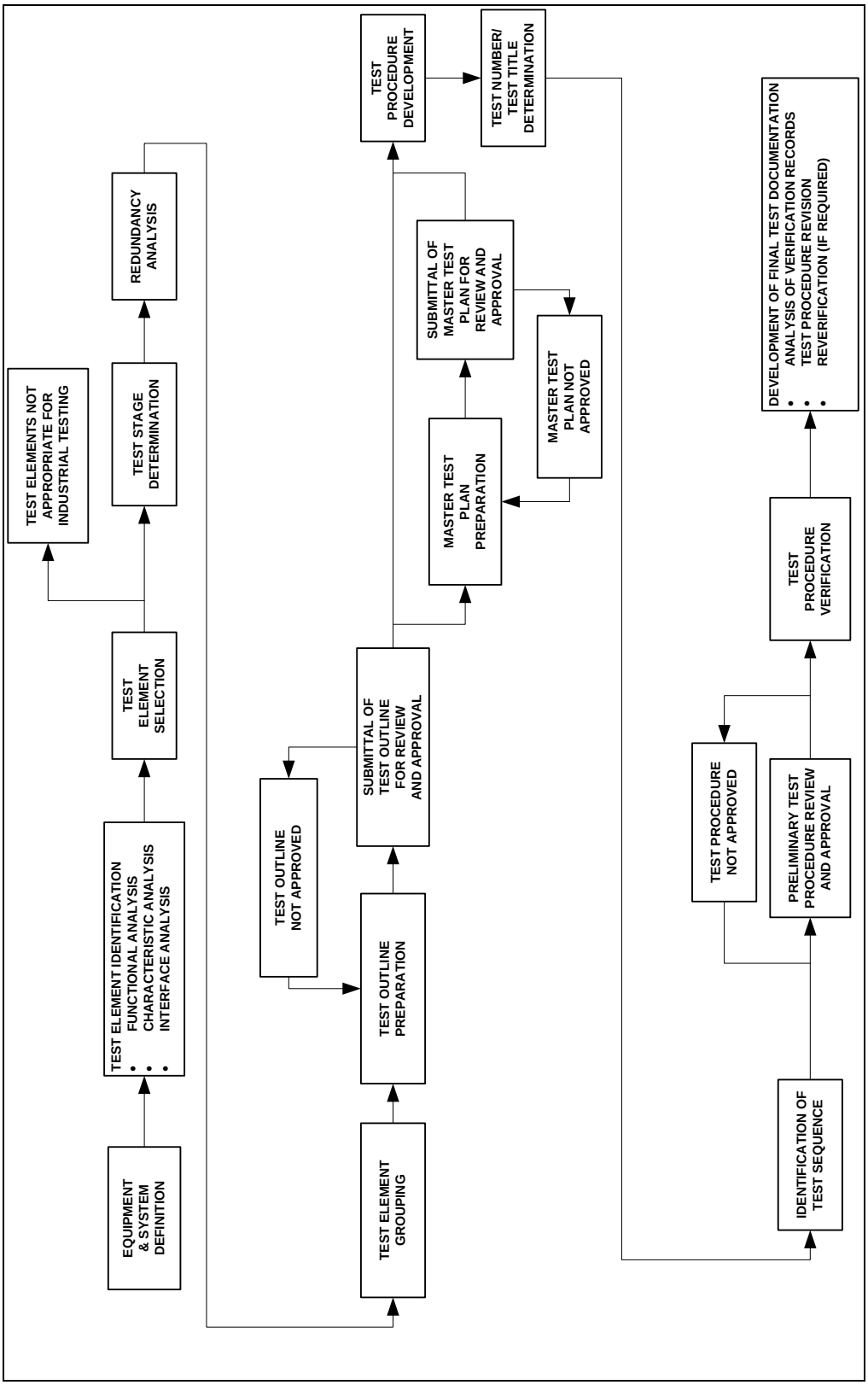


FIGURE 2. System engineering process for test documentation development.

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FUNCTIONS OF SURFACE SEARCH RADAR SYSTEM

1. Detect medium and large cross-section surface targets to line-of-site ranges.
2. Detect small targets at close ranges.
3. Provide radar video data to fire control system.
4. Provide radar video data to collision avoidance system.

FIGURE 3. Functional analysis results for surface search radar.

SURFACE SEARCH RADAR SYSTEM CHARACTERISTICS

TRANSMITTER:
- Peak power
- Pulse Width P PRF
- Pulse rise-time

RECEIVER:
- Sensitivity
- Noise Figure
- Bandwidth

ANTENNA:
- VSWR
- Gain
- Bandwidth
- Side lobe level
- Rotation rate
- Polarization
- Alignment

FIGURE 4. Characteristic analysis results for surface search radar system.

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TEST NUMBER/TEST STAGE/TEST TITLE DEFINITIONS

A.1 SCOPE

A.1.1 Scope. This appendix defines Test Number/Test Stage/Test Title definitions. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

A.1.2 Application. These Test Number/Test Stage/Test Title Definitions apply to all testing of shipboard equipment and systems performed in a shipyard or at sea during industrial periods.

A.2 DETAILED REQUIREMENTS - TEST NUMBERING

A.2.1 Test numbering procedures. Industrial Test Numbering is defined in the following paragraphs.

A.2.1.1 Introduction.

a. Policy and procedures for test numbering determination and assignment are described in the following sub-sections. As agents of NAVSEA 05E1, the official assignment of test numbers shall be performed by the Port Hueneme Division, Naval Surface Warfare Center (PHD NSWC) Code A34, using the Test and Evaluation Automated Management Information System (TEAMIS).

b. The authorized procedure for numbering and identification of test documents is based on the Expanded Ship Work Breakdown Structure (ESWBS). The master default ESWBS file list can be accessed, viewed, and retrieved in the reference section of Configuration Data Manager's Database-Open Architecture (CDMD-OA) and Configuration Management Reference Material (CMRM) websites www.cm.navsea.navy.mil and www.cm.navsea.navy.mil/cm/index.nsf, respectively. CDMD-OA and CMRM require user accounts. TEAMIS also provides an ESWBS quick reference list.

c. The test numbering policy and procedures provided herein enable standardization and control of the application of testing in the ship acceptance test environment. This process enables collection of test document information and consistent assignment of test numbers which will facilitate development, identification, storage, retrieval, and distribution of test documents. Numbered test documents include test outlines, test procedures, test summaries, and test reports. Such integrated test documents carry identical test numbers for the associated test requirement.

A.2.1.2 Policy. Requirements for the determination and assignment of test numbers are applicable to all military, government, and civilian organizations that develop test documents for use in Navy ship acceptance test programs. This applicability extends to test documents developed and/or used at land-based test facilities where test procedure verification is performed in advance of actual ship acceptance testing.

A.2.1.2.1 Test numbering requirement.

a. Each newly-developed test document shall be numbered in accordance with the requirements of this test numbering procedure.

b. Each revised test document shall be numbered (or renumbered) in accordance with the requirements of this section.

c. Test documents whose number does not comply with NAVSEA's current test numbering policy must have new test numbers assigned before being entered into the TEAMIS Repository.

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A.2.1.2.2 Test number format.

a. Test numbers shall contain nine digits exclusive of dashes, change, and revision characters. Test numbers shall be structured in accordance with the format depicted on figure A-1.

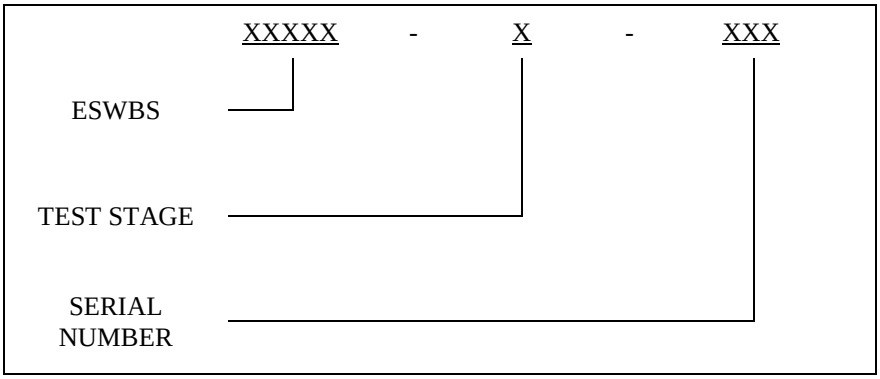


FIGURE A-1. Test number format.

b. The first five digits of the test number shall be based on ESWBS and shall comply with the master HSC cross reference (see A.2.1.b).

c. The sixth digit of the test number, the test stage, shall be determined and assigned based on the definitions in section A.3.

d. The last three digits of the test number shall be a serial number which is assigned to the respective ESWBS and test stage combination. The serial number digits shall be sequentially assigned based on review from TEAMIS Repository for the next available number, and shall not represent prerequisite testing relationships within any given test package.

e. Revisions and changes: Test numbers for existing test procedures may be modified by revisions (REV) or changes (CHG), if those changes to the test procedure (TP) have been formally approved by the test problem report (TPR) or test change proposal (TCP) process. Any such TP changes must be documented in the REVISION RECORD of the TP. TP changes that affect half or more of the TP pages or more are defined as Revisions; TP changes that affect less than half of the pages are a change (CHG). For example, the AN/SPS-49A(V)1 Initial Light Off test is TP number 45249-3-030. If the TP is updated in response to a TPR that affects more than half of the pages, the updated version of the TP would be 45249-3-030 REV A. If the revised version was then used on another ship and another TPR was submitted that required changes on less than half of the pages, the revised TP would then be 45249-3-030 REV A CHG 1.

A.2.1.2.3 Test number determination and request.

a. After development of the test document has begun and test stage determination has been completed, the Test Number/Test Title Request Form (see figure A-2) shall be initiated by the test development organization.

b. Responsible personnel within the test development organization shall determine the first six digits (ESWBS and test stage) of the technically correct test number. It shall be the responsibility of the TDA to determine a proposed test number based on that organization's knowledge of ESWBS, and on the test engineering performed to determine the scope of required testing and the appropriate test staging.

A.2.1.2.4 Test number assignment.

a. A single test number shall be assigned and issued for each test document related to a specific test requirement. The assigned test number shall be affixed to the test document.

b. The assigned test number shall be retained with the test document through each phase of development and throughout its life cycle.

c. After a test number has been assigned and issued, it shall not be used or re-used for any purpose other than that intended by this test numbering process.

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d. After the test number has been assigned it shall be entered, together with other required information, into the TEAMIS Repository.

A.2.1.2.5 Test number request.

a. When the test development organization has begun development or preparation of a new or revised test document, the request process should be initiated so that a record of such activity is available in the TEAMIS Repository.

b. TDAs shall review the Test Number/Test Title Request Form (see figure A-2) to determine data requirements for entry into the TEAMIS Repository. Test Number Requests may be submitted on line at the TEAMIS Web site by clicking on the “REQUEST NUMBER” button and completing the online form. Alternatively, the Test Number Request Form (see figure A-2) may be completed and e-mailed in PDF format to PRTH_NSWC_TEAMIS@navy.mil.

A.2.1.2.6 Test number assignment/verification. Responsibility for final test number assignment and entry into the TEAMIS test database rests with TEAMIS Management Operations, NSWC PHD Code A34. TEAMIS Management Operations will respond to all test number requests in writing verifying the official disposition of each test number request.

A.2.1.2.7 Test procedure submittal. Upon completion of the development or preparation of each new, revised, or re-numbered test procedure, the TDA will forward a copy for placement into the TEAMIS Repository. Test Procedures may be submitted in Microsoft Word or Adobe® Acrobat® Portable Document Format (PDF) file format, and should be submitted via e-mail to PRTH_NSWC_TEAMIS@navy.mil, or uploaded via the TEAMIS web site.

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TEST NUMBER/TEST TITLE REQUEST FORM

REQUESTING ORGANIZATION INFORMATION DATE OF REQUEST _____

1. Name _____
2. Organization and Code _____
3. Address _____
4. Phone: DSN _____ Com'l _____
5. TPMS Contact _____ Code _____
6. Purpose for TP Development
   ☐ new equipment/system ☐ modified equipment/system ☐ ship specific test procedure
   ☐ other _____

EQUIPMENT/SYSTEM INFORMATION

(Type/Model/Series)

1. Military Designation _____
2. Noun Name _____
3. Ship Class Applicability _____
4. Ship Change Document or other change information _____
5. ESWBS _____(five digits)
6. Stage _____(one digit)
7. Recommended Test Title (120 characteristics max) _____
8. Estimated Availability Date _____

Use reverse side of this form for additional comments.

TEST NUMBER/TITLE ASSIGNED (TO BE COMPLETED BY REPOSITORY)

Test Number

Test Title _____

Possible duplicate tests _____

FINAL REVIEW (TDD)_____ DATE_____

FIGURE A-2. Test number/test title request form (page 1).

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TEST NUMBER/TEST TITLE REQUEST FORM
(page 2)

Additional Comments:

FIGURE A-2. Test number/test title request form (page 2) – Continued.

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A.3 DETAILED REQUIREMENTS - TEST STAGE DEFINITIONS AND EXAMPLES

A.3.1 Scope. This appendix defines the seven stages of industrial testing and provides examples for stages two through seven. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

A.3.2 Application. These test stage definitions apply to all testing of shipboard equipment and systems performed in a shipyard or at sea during industrial periods.

A.3.3 Detailed requirements.

A.3.3.1 Test stages. Industrial testing is accomplished in seven stages as defined in the following paragraphs.

A.3.3.2 Stage one. material receipt inspection and shop tests. Stage one includes those tests and inspections that provide for inventory management and physical inspection of new material, equipment and systems, and associated documentation. These tests and inspections are intended to ensure receipt of equipment in good physical condition by the shipbuilder or other industrial organization. Stage one documentation is not normally in the form of a test procedure.

Stage one further includes those tests and inspections conducted prior to shipboard installation for new or repaired equipment and systems. In instances where equipment and systems are repaired aboard ship, shop test procedures may be used to validate readiness for shipboard testing. For work planning and cost accounting purposes, stage one is not a part of the test program and will normally be a part of the industrial organization's quality assurance program.

A.3.3.3 Stage two. shipboard installation inspections and tests. Stage two tests and inspections are conducted prior to operation of installed or relocated equipment, cabling, waveguide, piping, ventilation, etc., to ensure that each installation has been accomplished in accordance with established plans and specifications. The shipbuilder or industrial organization is normally responsible for the preparation of stage two test procedures.

A.3.3.4 Stage three. equipment tests. Stage three tests demonstrate that after shipboard installation, the individual equipment performs within the established limits and tolerances. These equipment operability tests are conducted independent of the system (i.e., the equipment may be isolated from the system) and can be conducted prior to complete system installation.

A.3.3.5 Stage four. intrasystem tests. Stage four tests demonstrate that equipment and required functions, entirely within one independent system, perform within established limits and tolerances. Stage four testing normally consists of intrasystem functions, signals, and commands within a single independent system of the combat system or ship system. Stage four includes all tests involving two or more items of equipment which do not involve more than one independent system of the combat system or ship system. Stage four tests may include tests between two or more items of equipment and between two groups of equipment within the same system.

A.3.3.6 Stage five. intersystem tests. Stage five tests involve two or more independent systems within the combat system, ship system, or between the combat system and the ship system. These tests demonstrate that two or more independent systems perform a specific function or functions within established standards. The exchange of intersystem signals, commands, functions, and all associated computer interfaces are included.

A.3.3.7 Stage six. special tests. Stage six tests require special simulation facilities or resources external to the immediate test organization, but are conducted as part of the dockside work package for the industrial effort. Special tests can apply to one or more items of equipment, a single system, or a number of systems, and may require total ship operability. Stage six tests that can only be performed at-sea should be designated as stage seven. Normally, there will be very few stage six tests in an industrial test program.

A.3.3.8 Stage seven. trials tests. Stage seven tests must be conducted during sea trials; e.g., Builders Trials (BT), Acceptance Trials (AT), Underway Trials (UT), and Post-Repair Trials (PRT). Test procedures are not identified with a stage seven number unless the test can only be conducted entirely or partially at sea.

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A.3.4 Test stage examples.

A.3.4.1 Stage two examples. The following are examples of stage two tests:

a. Cable integrity verification and insulation resistance checks.
b. Hydrostatic tests of piping systems.
c. Coolant flushing.
d. Lubrication of components before operation.
e. Inspect and clean air filters and air intakes.
f. Pre-energizing tests.
g. Reference planes and benchmark verification.

A.3.4.2 Stage three examples. The following are examples of stage three tests:

a. Equipment element level initial light off, diagnostic and functional tests.
b. Hinged/retractable mast operational test.
c. Machinery spaces lifting gear demonstration.
d. Shaft torsion meter performance test.

A.3.4.3 Stage four examples. The following are examples of stage four tests:

a. Intrasystem tests of two or more elements in the same system.
b. Propulsion seawater cooling system operational test.
c. Auxiliary propulsion system operational test.
d. Diesel generator parallel and reverse power load bank test.
e. System local area network interface test.
f. Navigation system alignment verification.
g. Sound powered telephone system operational test.

A.3.4.4 Stage five examples. The following are examples of stage five tests:

a. Intersystem interface tests.
b. Search radar distribution and display interface test.
c. SPS-73 external interfaces.
d. AN/SPQ-9B video and trigger verification.
e. AN/SPN-41A stabilization verification.
f. Integrated propulsion test.

A.3.4.5 Stage six examples. The following are examples of stage six tests:

a. Propulsion gas turbine engine removal and replacement demonstration.
b. EMI Phase I test (dockside).
c. HULL generated IMI test.
d. Data Link operational demonstrations dockside.
e. Navigation system certification (dockside).
f. SSDS MK 2 MOD 4B combat system integration test.

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A.3.4.6 Stage seven examples. The following are examples of stage seven tests:

a. Announcing system sound survey (at sea).

b. Radar set operational test (at sea).

c. Navigation system certification (at sea).

d. EMI Phase I test (at sea).

e. Underway airborne noise survey.

f. Hull vibration survey.

g. Landing system correlation verification (at sea).

h. Propulsion plant performance tests.

i. Link 4/11/16 operational demonstration (at sea).

j. SSDS MK 2 MOD 4B detect to engage demonstration (at sea).

A.4 DETAILED REQUIREMENTS - TEST TITLING DEFINITIONS

A.4.1 Test titling procedures. Industrial Test Titling is defined in the following paragraphs.

A.4.1.1 Introduction. Confusion in interpretation of the structure of test document titles has often resulted in unnecessary and costly development of new Navy acceptance test documents. With various organizations preparing test documents, and still others implementing those documents, a need for standardization in development and interpretation of titles has been identified. A lack of definition and standardization has also been recognized by test managers at different functional and managerial levels resulting in local attempts toward standardization. This procedure will promote standardization and consistency in development and assignment of test document titles generated by TDAs, and their subsequent interpretation by test managers and test program implementers. The principal purposes of test document titling guidelines are to:

a. Clarify and determine terminology to be used in the content of test document titles in order to reduce ambiguity, misinterpretation, and redundancy.

b. Provide standardization in the determination of titles to establish consistent understanding between TDAs and test document users.

c. Reduce the length of titles to essential minimums. Test titles may not exceed 120 characters, including spaces.

A.4.1.2 Policy. Requirements for the determination and assignment of test titles shall be applicable to all military, government, and civilian organizations that develop test documents for use in Navy ship acceptance test programs. This applicability extends to test documents developed and/or used at land-based test facilities where test procedure verification is performed in advance of actual ship acceptance testing.

A.4.2 Implementation procedure.

A.4.2.1 Test titling requirements. Requirements of this procedure shall be implemented in order to collect the information necessary for entry into the TEAMIS Repository and, subsequently, for management of test documentation by various responsible organizations.

a. Each newly developed or revised test document shall be titled in accordance with the requirements of this section.

b. Test titles shall be determined together with the test number and shall be finalized at the time of official test number assignment. Once a test title (and its associated test number) have been officially assigned and entered into the TEAMIS Repository, an approved Test Change Proposal (TCP) or Test Problem Report (TPR) shall be necessary to formally document any subsequent changes.

c. Existing test document titles need not be changed unless the test document is being revised. When a test document is being revised, the title shall be modified as appropriate as part of the revision process.

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d. Existing test documents may be re-titled at any time to comply with current test titling policy; however, an approved TCP or TPR shall be required to formally document the change.

A.4.2.2 Test title format and content.

a. Test document titles shall be limited to a maximum of 120 characters including spaces and hyphens. The use of symbols shall be avoided to reduce misinterpretation.

b. The inclusion of multiple test stages in a single test document shall be avoided to prevent misunderstanding document content and possible redundancy with other test documents. When, out of necessity, a test document must describe a test which covers two stages, the test title (and test number) shall always denote the higher stage test.

c. Each test document title shall be constructed in two to four sequential segments as shown on figure A-3. Segment 1 (Primary Hardware Identifier) and Segment 4 (Test Type) are mandatory in the title of each document. Segment 2 is used to identify interfacing equipment/systems at test stages four and five. Segment 3 is used to modify test types. Test titles shall contain no more than 120 characters.

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Primary Hardware Identifier	Secondary Hardware Identifier	Test Type Modifier	Test Type
(Segment 1)	(Segment 2)	(Segment 3)	(Segment 4)

Primary hardware identifier (segment 1). The Primary Hardware Identifier is mandatory in every test document title and shall consist of the AN/XXX-YY (V) or MK/MOD nomenclature when the associated hardware has such nomenclature. When AN or MK/MOD nomenclature has not been assigned, the official name of the equipment or system shall be used as the Primary Hardware Identifier. Use of the words system, subsystem, equipment, component, etc., shall be avoided because of the ambiguity in interpretation of their meaning. Also, these words require character spaces in the title fields which can be better utilized for other descriptive material. When AN or MK/MOD nomenclature is assigned, the full descriptive name shall not be used in favor of an accepted acronym. Examples of Primary Hardware Identifiers are:

a. MK 57 MOD 12 NSSMS
b. AN/SPS-49 RADAR

Secondary hardware identifier (segment 2). The Secondary Hardware Identifier shall be used to identify an interfacing item of equipment or a secondary system which is included in the test. Hence, Secondary Hardware Identifiers are only used with test stage four (intrasystem) and test stage five (intersystem) tests. Where both Primary and Secondary Hardware Identifiers are used, the hardware which is predominant or controlling the test shall be listed first. Examples of Secondary Hardware Identifiers (underlined) in conjunction with Primary Hardware Identifiers are:

a. AN/WSN-7 TO AN/SSN-6 NAVSSI
b. MK 23 MOD 1 TAS TO MK57 MOD10 NSSMS

Test type modifier (segment 3). The Test Type Modifier shall be used to qualify or augment the test type. Not all tests will require the use of a modifier. Examples of Test Type Modifiers (underlined) are provided below:

a. SIGNAL DATA PROCESSOR OFF-LINE DIAGNOSTIC
b. MK 15 MOD 22 CIWS GUN BARREL ALIGNMENT
c. ELECTRONIC DRY AIR UNOBSTRUCTED FLOW
d. REFRIGERATION PLANT PRESSURE TIGHTNESS
e. PROPELLER SHAFT BEARING CLEARANCE

Test type (segment 4). The Test Type states the functional objective which the test is intended to satisfy and is mandatory within each test document title. Examples of test titles using Test Types (underlined) are:

a. MK15 MOD 22 CIWS INITIAL LIGHT OFF
b. CATHODIC PROTECTION SYSTEM INSULATION RESISTANCE TEST
c. WASTE FOOD DISPOSAL OPERATIONAL TEST
d. AN/SPS-73 EXTERNAL INTERFACES
e. AN/SPS-49 TRANSMIT PERFORMANCE VERIFICATION
f. GENERAL ANNOUNCING SYSTEM SOUND SURVEY
g. ELECTRONIC DRY AIR UNOBSTRUCTED FLOW

FIGURE A-3. Test type segment identification.

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A.4.2.3 Test title determination and request.

a. The responsible personnel within the test development organization shall determine a proposed technically correct test title in accordance with this procedure. The recommended test title shall be requested, via Test Number/Test Title Request Form (figure A-2), concurrently with the test document number.

b. Once a test title has been assigned and entered into the TEAMIS Repository, it shall not be changed unless a TCP is prepared and submitted with appropriate approval signatures.

c. The assigned test title shall be affixed to the test document in accordance with the format requirements of Appendix B.

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FORMAT AND CONTENT FOR TEST OUTLINES, TEST PROCEDURES, TEST PROCEDURE APPENDICES, AND TEST PROCEDURE ADDENDUMS

B.1. SCOPE

B.1.1 Scope. This appendix establishes a uniform format for test outlines and test procedures and it describes information to be developed and provided. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

B.1.2 Application. This appendix is mandatory for test outline and test procedure preparation (see 5.2.7 and 5.2.9).

B.1.3 Warnings, cautions, and notes. Warnings, cautions, and notes shall be used as adjuncts to the text and are defined as follows:

WARNING: A warning is a statement used to call attention to possible personnel injury or loss of human life if the safety precaution is not observed. Warnings shall be listed immediately preceding the instructions in the test procedure.

CAUTION: A caution is a statement used to call attention to possible equipment damage. Cautions shall be listed immediately preceding the instructions in the test procedure.

NOTE: A note is a description of an unusual procedure or condition which must be brought to the operator's attention. Notes shall also be used to supply needed additional information. These shall not be steps in the test procedure. Notes may precede or follow the test procedure instruction.

B.1.4 Basic requirements. There are several components (sections) required for each test outline and test procedure. Appendices and addendums may be added to provide supplementary information. The primary components of a test procedure are listed as follows:

a. COVER SHEET
b. REVISION RECORD
c. LIST OF EFFECTIVE PAGES
d. TABLE OF CONTENTS (optional)
e. GENERAL INFORMATION
f. SAFETY INSTRUCTIONS
g. INITIAL CONDITIONS AND SETUP
h. TESTING STEPS
i. SHUTDOWN AND SECURING
j. TEST DATA RECORDING
k. TEST EQUIPMENT USED
l. COMMENTS

Each of the above sections shall begin on a separate page and contain a header for entry of the Test Title, Test No., and REV/CHG. In addition, the Data Sheet, Test Equipment Serial Numbers And Calibration Dates, and Comment Sheet pages shall have footers at the bottom of each page for entry of the Ship Hull No., Test Conductor Signature, Government Witness Signature, and Date. Detailed requirements for each Test Procedure section are provided on the following pages.

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B.2 REQUIREMENTS FOR TEST OUTLINES

B.2.1 Test outlines. Test outlines provide the information necessary for advance planning for test completion. Test Outlines include the COVER SHEET, REVISION RECORD, LIST OF EFFECTIVE PAGES, and GENERAL INFORMATION pages, and each of those sections should begin on a separate page, with the section title bold lettering at the top of each page bordered by parallel lines. The Test Outline becomes the front matter for the associated test procedure.

B.2.2 General information. The GENERAL INFORMATION pages include the following detailed information required for advance planning and conduct the test:

a. Test objective
b. Estimated testing time
c. References
d. Test equipment and support material requirements
e. Computer programs required
f. Prerequisite testing requirements
g. Special conditions and services
h. Equipment involved in the test
i. Method statement and station assignments

B.2.3 Test outline sample pages. Test Outline sections and data elements are illustrated in figures B-2 through B-9 on the following pages. To facilitate illustration of the examples, information that is not part of the test procedure, but is provided for amplifying information, will be enclosed in brackets and in Times New Roman font, or listed in a text box with Times New Roman font to distinguish it from the information that is part of the Test Procedure (see figure B-1).

SAMPLE AMPLIFYING INFORMATION TEXT BOX

Information provided on the following pages in text boxes is not part of the test procedure, but is provided for amplifying information and illustrative purposes.

FIGURE B-1. Amplifying information.

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TEST TITLE: TEST NO:
REV/CHG:

COVER SHEET

Prepared by: _____

Name

_____ _____

Organization/Code Date

Reviewed by: _____

Name

_____ _____

Organization/Code Date

Approved by: _____

Name

_____ _____

Organization/Code Date

The test document preparer shall provide a cover sheet to include the indicated information.

The cover sheet may be modified to include additional information appropriate for the industrial test program in which the test procedure will be utilized.

FIGURE B-2. Cover sheet.

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TEST TITLE: TEST NO:
REV/CHG:

REVISION RECORD

REV/CHG	DESCRIPTION	APPROVAL INITIAL	APPROVAL DATE

LIST OF EFFECTIVE PAGES

PG-REV	PG-REV	PG-REV	PG-REV	PG-REV	PG-REV	PG-REV

Starting at top left, insert the page number, dash, revision/change. Fill in each column from left to right. Original pages will have nothing following the dash. Include extra pages if needed.

Sample data:

1-B	2-	3-B	4-B	5-B	6-B	7-C
8-B	9-A/1	10-	11-C	12-B	13-B	14-B
15-C	16-B	17-C	18-C	19-C	20-C	21-C
22-C	23-C	24-B	25-C	26-C	27-B	28-

FIGURE B-3. Revision record and list of effective pages.

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TEST TITLE: TEST NO:
REV/CHG:

TABLE OF CONTENTS

SECTION	TITLE	PAGE
A	GENERAL	6
B	STATION ASSIGNMENTS	10
C	INITIAL CONDITIONS AND SETUP	11
D	PROCEDURE	14
	POWER SUPPLY AND LOGIC CIRCUITS TEST	14
	TELEPHONE, TRAIN WARNING BELL, AND GUN HOUSE LIGHTS TEST	27
	DISPLAY PANEL EP3 TEST	29
	STATIC BLEED TEST	30
	CHECK UPPER ACCUMULATOR MOTOR BPX1 FOR PROPER ROTATION	51
	CHECK LOWER ACCUMULATOR MOTOR BPX2 FOR PROPER ROTATION	52
	FLUSH FUZE SETTER	52
	INSTALL ELECTROHYDRAULIC SERVO VALVE	53
	CHECK UPPER ACCUMULATOR EMERGENCY MOTOR BPX3 AND LOWER ACCUMULATOR EMERGENCY MOTOR BPX4 FOR PROPER ROTATION	54
	CHECK BLOWER MOTOR BPZ1 FOR PROPER ROTATION	55
	LOADER MOTORS INTERLOCKS TEST	55
	TRAIN AND ELEVATION MOTOR ROTATION AND INTERLOCKS TEST	59
	CHECK EP4 (ELEVATION) AND EP5 (TRAIN) DISPLAY PANELS	62
	CHECK TEST CONTROL PANEL LIGHTS AND DISPLAYS	62
E	SHUTDOWN	64
F	DATA SHEET	65
	MT51	65
	MT52	81
G	TEST EQUIPMENT SERIAL NUMBERS AND CALIBRATION DATES	97
H	COMMENT SHEET	98

TABLE	TABLE	PAGE
1	EP1 TURN-ON	12
2	EP2 TURN-ON	13

FIGURE B-4. Sample table of contents page 1.

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TEST TITLE: TEST NO:
REV/CHG:

TABLE OF CONTENTS

TABLE	TABLE	PAGE
3	EP2 TURN-ON (cont.)	13
4	TEST PLATE	14
5	BLEED VENT PLUGS	31
6	BLEED PIPES (2 sheets)	32-33
7	EQUIPMENT VENT PLUGS	42
8	EQUIPMENT VENT PLUGS	43
9	BLEED COMPONENTS	44
10	BLEED COMPONENTS	49

FIGURE	FIGURE	PAGE
1	LOWER ACCUMULATOR SYSTEM BLEED POINTS	34
2	LOWER ACCUMULATOR SYSTEM BLEED POINTS	35
3	LOWER ACCUMULATOR SYSTEM BLEED POINTS	36
4	LOWER ACCUMULATOR SYSTEM BLEED POINTS	37
5	LOWER ACCUMULATOR SYSTEM BLEED POINTS	38
6	LOWER ACCUMULATOR SYSTEM BLEED POINTS	39
7	LOWER ACCUMULATOR SYSTEM BLEED POINTS	40
8	LOWER ACCUMULATOR SYSTEM BLEED POINTS	41
9	UPPER ACCUMULATOR SYSTEM BLEED POINTS	45
10	UPPER ACCUMULATOR SYSTEM BLEED POINTS	46
11	UPPER ACCUMULATOR SYSTEM BLEED POINTS	47
12	TRAY ASSY	48
13	CRADLE TO SLIDE LATCH PIPING	49
14	LOWER HOIST COMPONENTS BLEED POINTS	50

FIGURE B-5. Sample table of contents page 2.

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TEST TITLE: TEST NO:
REV/CHG:

GENERAL INFORMATION

1. OBJECTIVE: [Briefly describe the test objective relative to capabilities and/or interfaces to be tested.]

2. ESTIMATED TESTING TIME:

[Enter the estimated elapsed (clock) time in hours required to perform the test, assuming no interruptions. Includes the time required to complete the INITIAL CONDITIONS AND SETUP and SHUTDOWN AND SECURING steps.]

3. REFERENCES:

[List the identifying number and title of reference sources used as a primary basis for preparation of the test outline and procedure.]

4. TEST OR SUPPORT EQUIPMENT AND MATERIAL:

GENERIC NAME	QUANTITY	IDENTIFYING INFORMATION

[List all portable, non-installed test or support equipment and material including special tools, handling equipment, measurement devices, shapes, etc., required to conduct the test. Common tools should not be listed. Standard shipboard test equipment as defined in NAVSEA ST000-AA-IDX-010-PEETE should be used wherever possible.]

[List electrical and electronic test equipment by generic name including the subcategory (SCAT) code under IDENTIFYING INFORMATION. If test equipment does not have SCAT code, list the manufacturer and model number, drawing number, or other identifying information. For nonelectrical and electronic test equipment, list by generic name and provide salient characteristic in IDENTIFYING INFORMATION column or manufacturer's part number and Federal Supply Code for Manufacturer (FSCM) as listed in Cataloging Handbook H4. Special tools and materials will be identified by name, manufacturer's part number, and FSCM.]

5. COMPUTER PROGRAMS REQUIRED:

[List all computer programs required in the conduct of the test and identify all test data media (tapes, CDs, etc.). The level of detail to which computer programs are specified will vary depending on requirements. In some cases the version number, installed patches, and date may be required. However, it is advisable to specify computer programs without versions if not absolutely required. Specifying specific program version numbers will cause test procedures to be out of date if/when program versions are updated/changed (i.e., SSDS MK2 MOD 2 Operational Program vice SSD MK 2 MOD 2 OP Program version 5.04.07).]

Note: In all cases, software provided for testing purposes only must be removed before the ship is delivered. Upon ship delivery, all installed software must be as certified and approved for use in accordance with the NAVSEAINST 9410 series.

6. PREREQUISITES:

[List, by test number and title, the immediate prerequisite test(s) that must be completed prior to the conduct of this test. Do not require a complete test if only a portion of that test is required. In such cases, prerequisite functions should be called out by listing the specific steps in the prerequisite test procedure.]

FIGURE B-6. General information format and content (sheet 1 of 2).

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TEST TITLE: TEST NO:
REV/CHG:

GENERAL INFORMATION

7. SPECIAL CONDITIONS AND SERVICES:

[Identify any special or unique support services, facilities, or conditions required for test conduct; e.g., air, water, electrical services, dry-dock, and support ships or aircraft. Include detailed requirements for services.]

8. EQUIPMENT INVOLVED IN TEST:

[List all shipboard installed equipment involved in the test. When a complete system is involved, the system is listed and not the individual components; e.g., AN/SPS-48E. When more than one type of equipment may be used, identify alternatives and use generic name in TESTING STEPS. For example, if gyro heading is obtained from an AN/WSN-5 on one ship and from the AN/WSN-7 on another ship, both alternatives should be listed and testing steps should refer to "Ships Heading" rather than "WSN-7 Output."]

9. CONFIGURATION:

[List specific system configurations that the test procedure was engineered to support, or APL number(s) of the equipment or system to which the test applies.]

10. METHOD:

[Explain briefly in narrative form the scope of the test and generally how the test is to be performed. Include functions tested, modes of operation of major equipment, intended use of test equipment, special support services, and facilities. Avoid redundancy with the test objective paragraph.]

[Test method section should be brief and not include references to other documents. The method may be divided into subtests for convenience and clarity.]

11. STATION ASSIGNMENTS:

STATION	NO. PERSONNEL	COMMENTS

[List all stations by the component name that must be manned to perform the test. Indicate the number of personnel including the test conductor, and each technical discipline (e.g., electrician, pipefitter, electronic technician) required at each station. Station assignments should provide an indication of the total number of personnel required to conduct the test; therefore, if one person can reasonably man more than one station, this should be clearly indicated.]

12. SUPPLEMENTAL INFORMATION:

[If additional administrative information would be helpful to conduct of the test, such as a list of abbreviations, or a list of definitions/glossary information for key technical terms, that information may be added to the general information as paragraph "12. Supplemental Information". Equipment set up information (for example, tables of settings for dip switches), should not be listed here, but rather should be provided in the Initial Conditions and Setup pages or Test Step pages, or may be added as an Appendix at the end of the test procedure. See B.3.3 for details on test procedure appendices and addendums.]

FIGURE B-7. General information format and content (sheet 2 of 2).

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TEST TITLE: TEST NO:
REV/CHG:

GENERAL INFORMATION

12. SUPPLEMENTAL INFORMATION:

CONVENTION USED:

a. Word(s)/character(s) enclosed in < > indicates a button, switch, VAB, or Window button or icon that must be pushed, pressed, repositioned or clicked on. For example:

- Set <AUTO/MANUAL> indicator switch to **AUTO**.
- Verify <RADAR STATE> rotary switch is set to **RADIATE**.
- In Interface Control Window, toggle <Enable/Disable> to display **Disable/Green Up-Arrow**.

b. Word(s)/character(s) enclosed in [ ] Indicate a light, indicator, field, or label to be observed. For example:

- Observe [READY] is **lit**.
- At System Status Window, verify [NAVSSI] is **Green**.

c. **Bolded** words is the position or condition a button, switch, VAB, or Window button or icon is to be selected or place in; the state of a light, indicator; or the data in a field or label to be observed as illustrated in the above examples.

d. Word(s)/character(s) enclosed by { }: Indicates the word(s)/character(s) must be typed at a data entry device. For example:

- {pingAppLayer "193.2.1.2",5} means type: pingAppLayer "193.2.1.2",5
- At login prompt enter {password} means type: password
- {CTRL+ALT+DEL} means depress: CTRL, ALT, and DEL keys simultaneously.

Note that all characters inside the brackets {} should be typed, including the quotation (") character in the above example, but the brackets themselves are not typed.

e. Word(s) in ***"bold italics"*** indicates an alert, message or display, usually resulting from an immediately preceding operator action.

f. Bearing and ranges are from Ownship, unless otherwise stated.

g. Scaling for ranges will be indicated as follows:

- Ranges in nautical miles – nm
- Ranges in yards - yds or Kyds

This is a good example of how to use a Supplemental Information paragraph in the Test Outline to explain testing conventions.

FIGURE B-8. General information supplemental information sample 1.

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TEST TITLE: TEST NO:
REV/CHG:

GENERAL INFORMATION

12. SUPPLEMENTAL INFORMATION:

ABBREVIATIONS:

NOTE: Following abbreviations may be used within this procedure:

ABRSP	Auxiliary Batteries Release Switch Panel (SSDS)
ADWC	Air Defense Warfare Coordinator (SSDS operator position)
AECM	Active Electronic Countermeasures
AIMS	ATCRBS, IFF Mk 12 System
AM	Alert Manager (SSDS software component)
ASTAB	Automated Status Board
ATC	Air Traffic Control
ATP	Auxiliary Touch Panel
BFTT	Battle Force Tactical Training (AN/USQ-T46)
BOPC	BFTT Operator Processor Console
CCG	Command Control Group (OK-644/S or OJ-774/UYQ-70)
CDM	CEC ID Doctrine Manager (SSDS software component)
CDS	Combat Direction System
CEC	Cooperative Engagement Capability
CEP	Cooperative Engagement Processor
CEPIN	CEP Integration (SSDS software component)
CETPS	Cooperative Engagement Transmission Processing Set (AN/USG-2)
CFAR	Constant False Alarm Rate
CIC	Combat Information Center
CRS	Canister Round Simulator (RAM)
CSOSS	Combat System Operational Sequencing System
CT	Command Table
CTA	CEC Training Adjunct
CTSL	Common Track Store Locator
CU	Cooperating Unit (CEC)
DCS	Data Collection System (BFTT)
DCS	Display & Control Subsystem (NAVSSI)
DDCN	Data Distribution and Control Network
DDS	Data Distribution System (CEC)
DPG	Data Processing Group (OL-657/S)
DSVL	Doppler Sonar Velocity Log (AN/WQN-2)
EA	Electronic Attack
ESM	Electronic Support Measures (AN/SLQ-32)
EWSUP	Electronic Warfare Supervisor (SSDS operator position)

This is an example of how Test Outline paragraph 12 – Supplemental Information – can be used to provide a glossary of terms used in the Test Procedure.

FIGURE B-9. General information supplemental information sample 2.

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B.3 REQUIREMENTS FOR TEST PROCEDURES

B.3.1 Test procedure content. Test procedures consist of the same COVER SHEET, REVISION RECORD, LIST OF EFFECTIVE PAGES, and GENERAL INFORMATION that are provided as the "test outline". Test procedures shall contain the appropriate steps and information necessary to accomplish the test objectives by the methods described in Test Outline section. Test procedures shall be written to ensure that the procedural steps to be followed are clearly understood by the test conductor. If supplementary information from technical manuals is needed, it should be provided as an appendix or addendum that is included as part of the test procedure. Test procedures shall be non-redundant and not require external documentation or supplementary information, with the exception of classified supplements provided as addendums. The body of each test procedure shall consist of SAFETY INSTRUCTIONS, INITIAL CONDITIONS AND SETUP, TESTING STEPS, SHUTDOWN AND SECURING, TEST DATA RECORDING, TEST EQUIPMENT USED, COMMENTS, APPENDICES, and ADDENDUMS (if any). Each of those sections shall begin on a separate page. Exceptions to pagination requirements may be granted for extremely brief Test Procedures of three pages or less. In those cases, the following sections shall begin on a separate page: COVER SHEET, GENERAL INFORMATION, SAFETY INSTRUCTIONS, and TEST DATA RECORDING.

B.3.2 Headers and footers. Each page of the test procedure shall have a header for entry of the Test Title, Test Number, and REV/CHG. In addition, the TEST DATA RECORDING, TEST EQUIPMENT USED, and COMMENTS pages shall have footers at the bottom of each page for entry of the Ship Hull Number, Test Conductor Signature, Government Witness Signature, and Date.

B.3.3 Appendices and addendums.

B.3.3.1 Appendix content. Appendices are used to provide common information applicable to test procedures, and to avoid repetition of the same steps in individual test procedures. Examples are turn-on/turn-off procedures or system/sub-system initialization procedures. Appendices shall be included as part of the Test Procedure if they are referenced by the Test Procedure, after the final COMMENTS page. The appendices will be identified in the table of contents by a letter designator, e.g., Appendix A, Appendix B, etc. Appendix page numbers will be numbered A-1, A-2, etc. (using the appendices' letter designator for the alphabetic character). The following components (sections) and sequence represent the minimum content for all turn-on/turn-off and initialization appendices and each of the following sections shall begin on a separate page:

a. Appendix cover sheet
b. Revision record
c. Table of contents (optional)
d. List of effective pages
e. Procedure steps and instructions

B.3.3.2 Addendum content. Test procedure addendums are supplements that provide the classified portions of a test procedure, such as performance or design specification values, parameters, frequencies, codes, etc. Addendums will be identified in the table of contents and numbered as described above for appendices. However, the purpose of the addendum is to include classified information. Therefore, addendums must be kept separate from the Test Procedure. An unclassified addendum page should be provided in the Test Procedure that identifies the need for the classified supplement, and provide detailed instructions to obtain the classified addendum. A test procedure addendum shall have as a minimum the following components:

a. Cover sheet
b. Revision record
c. List of effective pages
d. Classified data, instructions or procedures

B.3.4 Test procedure sample pages. Test procedure sections and data elements are illustrated in figures B-10 through B-19 on the following pages. Information that is not part of the Test Procedure, but is provided for amplifying information, will be enclosed in brackets and in Times New Roman font, or listed in a text box with Times New Roman font to distinguish it from the information that is part of the Test Procedure.

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TEST TITLE: TEST NO:
REV/CHG:

SAFETY INSTRUCTIONS

[Describe the safety factors that will impact test implementation and the requirements that must be met prior to and during test conduct. Instructions will include safeguards necessary to prevent injury to test personnel and observers, and to prevent damage to equipment under test. Examples are special rigging, rubber deck matting, equipment, procedures, and warnings to be posted.]

FIGURE B-10. Safety instructions format and content.

TEST TITLE: TEST NO:
REV/CHG:

INITIAL CONDITIONS AND SETUP

STEP	STATION	INSTRUCTION

[Provide the necessary steps to establish the initial conditions and equipment setups required to start the test procedure. Appendices that contain standard turn on procedures may be provided as part of the test procedure. The test procedure shall be structured so that no technical data is recorded from the steps in this section. All technical data shall be recorded based on steps contained in the procedure. The information contained in this section is developed in three columns:

a. STEP. Steps shall be sequentially numbered starting with step 1. Sub-headings shall not be numbered. Only operator actions are numbered. As a general rule, for each complete action performed or observation made, a separate step shall be used. Sub-steps may be used when intermediate actions are necessary to perform a complete action. They will be indicated by lower case alpha characters placed in the instruction column.

b. STATION. The appropriate location of the equipment or operator where the action in the instruction is taking place. This includes the location and equipment nomenclature (equipment short title is acceptable) of the station. Subsequent consecutive actions performed on the same unit do not require a station entry.

c. INSTRUCTION. The detailed step-by-step instructions for bringing the equipment under test to a predetermined condition. The system is normally considered to be “powered off.” Unnumbered sub-headings may be used to group or identify steps to accomplish an intermediate objective. It is desired that all text be left justified; however, indentation can be used if clarity or cautions/warnings need highlighting.]

FIGURE B-11. Initial conditions and setup format and content.

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TEST TITLE: TEST NO:

REV/CHG:

TESTING STEPS

STEP STATION INSTRUCTION

[TESTING STEPS shall be formatted in three columns: STEP, STATION, and INSTRUCTION.]

a. STEP. Steps shall be sequentially numbered starting with step 1. Sub-headings shall not be numbered. Only operator actions are numbered. As a general rule, for each complete action performed or observation made, a separate step shall be used. Sub-steps may be used when intermediate actions are necessary to perform a complete action. They will be indicated by lower case alpha characters placed in the instruction column.

b. STATION. The appropriate location of the equipment or operator where the action in the instruction is taking place. This includes the location and equipment nomenclature (equipment short title is acceptable) of the station. Subsequent consecutive actions performed on the same unit do not require a station entry.

c. INSTRUCTION. Provide the detailed step-by-step instructions necessary to accomplish the objectives of the test. Statements shall be concise and clear as to what the test operator must do or observe and, whenever possible, begin with an action verb. Any test function/subtest must be called out with an underlined header. Unnumbered sub-headings may be used to group or identify steps to accomplish an intermediate objective. It is desired that all text be left justified, however, indentation can be used if clarity or cautions/warnings need highlighting.

Safety “WARNINGS” and “CAUTIONS” shall precede the step or steps to which they apply and shall not be numbered. WARNINGS and CAUTIONS shall have the words “warning” and “caution” on a separate line, be capitalized, bolded, and underlined. NOTES shall not be numbered, but shall have a blank line before and after it. NOTES shall have the word Note bolded and underlined. Figures or tables shall be on the same page or the page immediately following the first reference to the figure or table. Figures are labeled using Arabic numerals and tables may be indicated by capital letters or Arabic numerals.

Steps describing the recording of data shall use the words “Record on data sheet.” All data taken during the test must be in response to a step contained in this section. When a test is conducted more than once to accommodate the testing of more than one of the same equipment, the serial number of the equipment under test shall be entered on the data sheet.

In cases where charts, photographs or computer printouts are required for analysis of data, the step shall indicate a data sheet entry of the pass/fail criteria. The actual charts, photos or printouts shall be retained by the designated local government representative such as the Test Director or Local Total Ship Test Director (LTSTD). A series of actions, which are repeated, may be referred to after the first time they are used provided it does not involve turning the page or where the expansion of the test through repeated writing out of the steps would make the procedure unwieldy.

FIGURE B-12. Testing steps format and content.

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TEST TITLE: TEST NO:
REV/CHG:

TESTING STEPS

STEP	STATION	INSTRUCTION
71	Loader Room	Remove goggles and rubber gloves.
		CAUTION ENSURE VALVES ARE POSITIONED SO NOT TO DAMAGE VALVES OR EQUIPMENT.
		WARNING ENSURE 440 VAC NORMAL & ALTERNATE IS OFF AND TAGGED OUT AT THE MBT AND EMERGENCY 440 IS OFF AND TAGGED OUT AT THE SWITCH PRIOR TO CORRECTING MOTOR ROTATION.
		NOTE: To ensure tag out procedures are performed only once, check all motors rotations then tag out 440 VAC and correct as necessary.
		NOTE: Do not remove flushing block or install train and elevation electrohydraulic servo valve.
		CHECK UPPER ACCUMULATOR MOTOR BPX1 FOR PROPER ROTATION
72	EP1/ EP2	Ensure 440 VOLTS POWER ON lights (three lights) are illuminated on the EP1. Set switches to positions shown on Table 1, Column L, Table 2, Column A from section C, Initial Conditions and Setup.
73	EP2/ Gun Mount	From the OPERATIONS page, select Motor Controls, touch Upper Accumulator-Motor Start/Off followed by Upper Accumulator-Motor Stop/Off touchbutton. Compare motor shaft rotation with rotation arrow on motor housing. Record on Data Sheet.

Above sample of testing steps illustrates how WARNINGS, CAUTIONS, and NOTES are inserted. This sample also illustrates how a sub-test (Check Upper Accumulator Motor) is identified by an underlined header.

FIGURE B-13. Testing steps sample 1.

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TEST TITLE: TEST NO:
REV/CHG:

TESTING STEPS

STEP	STATION	INSTRUCTION
		SECTION B: EW
9.	EW SUP	Manually associate ES track (if not automatically associated). A. <Pri Hook> the ES track (Brg 230) to be associated. B. <Sec Hook> the surface track (Brg 240) to be associated. C. At Primary Array, select <ASSOC> VAB. D. At the ES Close Control Window: 1. Observe the applicable data associated with the Primary hooked ES track is added to the data associated with the Secondary hooked track. 2. Observe the track number of the Primary hooked ES track is added to the ASTRK field of the Secondary hooked track, and the track is added to the EW Association Summary Window.
10.	EW SUP	Repeat step 9 for other candidate tracks.
11.	BOPC	Stop and reset scenario.
EA ENGAGEMENTS		
12.	TC	Confirm Combat Systems Training Configuration. Confirm or conduct steps to authorize PTT.
13.	SLQ	Place console in "AECM ON".
14.	EW SUP	Authorize EA. A. At Primary Array: 1. Toggle <EACM AUTH> to **YES**. 2. At confirmation prompt, select <CONFIRM>.
		EA ENGAGEMENTS FROM ADWC
15.	BOPC	Restart scenario (EW) at time 00:15:00.
16.	ADWC	Manually engage with EA. A. Hook hostile ES track at Brg 060. B. At Primary Array, select <ENGAGE>. C. At Weapon Select Array, select <EA>. D. At TACSIT, observe track symbol engagement modifier and pairing line.

This sample illustrates how sub-test areas are sequentially identified by letters and designated with an underlined header. This sample also illustrates how sub-steps are listed in the instructions with letters to identify each sub-step.

FIGURE B-14. Testing steps sample 2.

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TEST TITLE: TEST NO:
REV/CHG:

SHUTDOWN AND SECURING

STEP	STATION	INSTRUCTION

Provide all of the necessary information to turn off equipment, disconnect test equipment and services, and restore the equipment or system to pretest conditions.

Steps shall be sequentially numbered, starting with step 1.

Shutdown instructions shall not require gathering or recording of data. All data observation and recording should be part of the testing steps.

Shutdown instructions may not refer to technical manuals or other publications. However, appendices that contain standard shutdown procedures may be included in the Test Procedure and referenced in the shutdown and securing instructions.

Avoid generalized statements such as “Return equipment to normal operating condition.”

SHUTDOWN AND SECURING pages shall include a signature section as depicted below.

FIGURE B-15. Shutdown and securing format and contents.

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TEST TITLE: TEST NO:
REV/CHG:

TEST DATA RECORDING

TEST DATA RECORDING pages are used to record all data called out in testing steps for measurement or observation. In the event a test is performed more than once (i.e., more than one equipment of same nomenclature), a separate set of data sheets shall be included for each time the test is to be run. This section is not applicable to appendices and may not be applicable to an addendum. The information contained in this section is developed in four columns:

a. STEP. Steps shall be numbered to correspond with the appropriate PROCEDURE section step.

b. FUNCTION. The name of the function or parameter to be recorded shall be entered.

c. EXPECTED RESULTS. Entries of parameters which have definite values (yards, feet, volts, amps, etc.) shall show the nominal value followed by its tolerances, e.g., 10,000 yds +/- 10 yds, .01 V +/- .001 V, etc. Commonly accepted abbreviations should be used. Values shown with no tolerances shall be considered to require the exact value. Those steps involving the observation of illuminated indicators shall show illuminated (color where applicable) or extinguished. Those steps involving switch settings shall show the actual switch position, i.e., ON, #7, ACC, etc. Parameters involving observation of alerts or PPI displays shall show: satisfactory, observed, received, displayed, transmitted, etc. Under no circumstances should “SAT/UNSAT” be the expected results.

d. ACTUAL RESULTS. The actual results observed during test procedure conduct shall be recorded in this column. Sufficient space shall be provided to record expected results. A result must be recorded for each entry in the Expected Results column. For those entries in the Expected Results column requiring a specific value, the actual value will be recorded. For those entries in the Expected Results column requiring an observation, the entry of a check mark will indicate a satisfactory observation. In the case of an unsatisfactory observation, the word UNSAT will be entered under the Actual Result column and an explanation shall be entered on the COMMENTS sheet.

Note the signature section, which is a requirement on Test Data Recording sheets.

SHIP HULL NO.	TEST CONDUCTOR SIGNATURE	GOVERNMENT WITNESS SIGNATURE	DATE

FIGURE B-16. Test data recording format and content.

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TEST TITLE:

Note how this data sheet can be used for more than one system by using system serial numbers on each set of data recording sheets.

TEST NO:
REV/CHG:

TEST DATA RECORDING

GUIDED MISSILE FIRE CONTROL SYSTEM MK 57 NSSMS SERIAL NO. _____
GUIDED MISSILE LAUNCHING SYSTEM MK 29 SERIAL NO. _____

PREREQUISITES

48291-3-152 MK 29 MOD 3 Launcher Movement – DCU Control/Local Control (NSSMS MOD 10/11)
48291-4-131 MK 57 MOD 10/11 Tracker Illuminator Console/XMTR Group Interface
48291-4-133 MK 9 Radar Calibration and Operability (NSSMS MOD 10/11)
47282-3-001 AN/SSQ-82(V) Equipment Checkout

Prerequisites Completed _____ Signature/Date _____

STEP	TEST ELEMENT	EXPECTED RESULTS	ACTUAL RESULTS
2.	DIRECTOR BEARING	Air ready	_____
	DIRECTOR ELEVATION	Air ready	_____
3.	LAUNCHER TRAIN	Air ready	_____
	LAUNCHER ELEVATION	Air ready	_____
4.	BITE TEST CONTROLS-FAULT INDICATORS 1 through 12	Not lit	_____
	TEST TRAINING ON	Not lit	_____
	TEST TRAINING OFF	Lit	_____
	SAFE 1-4	Lit	_____
	SAFE 5-8	Lit	_____

SHIP HULL NO.	TEST CONDUCTOR SIGNATURE	GOVERNMENT WITNESS SIGNATURE	DATE

FIGURE B-17. Test data recording format and content.

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TEST TITLE: TEST NO:

REV/CHG:

TEST EQUIPMENT USED

List all test equipment utilized in the test, including all general and specialized test equipment, special test cables, attenuators, and any other materials requiring calibration. Include extra sheets as necessary to identify all test equipment.

GENERIC NAME	MODEL	SERIAL NO.	CALIBRATION DUE DATE	REMARKS

TEST EQUIPMENT USED documents the non-installed test equipment that is actually used during the conduct of the test procedure. Data is to be recorded from nameplate and calibration sticker. If the test spans several days and the test equipment is used on separate dates, an entry is required for each separate day the equipment is used.

This component is part of the Test Procedure, and shall not be used in appendices or addendums.

SHIP HULL NO.	TEST CONDUCTOR SIGNATURE	GOVERNMENT WITNESS SIGNATURE	DATE

FIGURE B-18. Test equipment used format and contents.

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TEST TITLE: TEST NO:

REV/CHG:

COMMENTS

This sheet is provided for the test conductor and government witness to make appropriate comments including the following:

a. Visual observation of dynamic responses;
b. Erratic or unusual equipment behavior;
c. Operational or handling difficulties;
d. Procedural corrections;
e. Equipment malfunctions;
f. Discrepancies noted during test conduct; and,
g. Waivers, including reference to authorization document; i.e., letter, message, etc.

Indicate if a Test Problem Report (TPR) was generated with respect to these or other problems.

The COMMENTS sheet is used by the test conductor and Government witness to record significant events, conditions, and observations pertinent to test conduct.

This section is not applicable to appendices or addendums.

The COMMENTS sheet will be used to indicate the numbers of any Test Problem Report (TPR), Software Trouble Report (STR) or Computer Program Problem Report (CPPR) and is prepared during the conduct of the test.

A line for recording equipment/system identification or serial number shall be included at the beginning of this section. Prerequisites as listed in GENERAL INFORMATION section 6 shall be listed here also with a signature block to verify completion of the prerequisites.

SHIP HULL NO.	TEST CONDUCTOR SIGNATURE	GOVERNMENT WITNESS SIGNATURE	DATE

FIGURE B-19. Comments format and content.

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TAILORING INSTRUCTIONS

C.1. SCOPE

C.1.1 Scope. This appendix provides for tailoring the test procedure development process to meet the needs of individual test programs. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

C.1.2 Applicability. There are circumstances in which the engineering process is inappropriate for development of test procedures as outlined in this document. In such cases the requirements of the standard may be tailored to conform to program constraints or to take advantage of economies afforded by previous development efforts.

C.2. DETAIL REQUIREMENTS

C.2.1 Tailoring philosophy. The requirements of this standard may be tailored to meet the needs of individual programs. However, any tailoring of the standard must be done in a way that will preserve the integrity of the development process. The development process prescribed by the standard has two objectives:

a. To develop test procedures that, as written, can be performed in a shipboard environment, and where any deviation of actual results from expected results can be attributed to equipment deficiencies.

b. To develop test outlines and test procedures that completely and adequately demonstrate the material readiness, operability and functionality of the equipment, or systems tested (i.e., ability of shipboard equipment and systems to perform their intended function).

Failure to verify the first objective becomes readily apparent during industrial testing and is reflected in the number of test problem reports and test change proposals submitted. Failure to meet the second objective results in the deployment of ships in an unknown and often inadequate state of readiness. The latter shortcomings are the more serious of the two and are the more difficult to detect and correct. Consequently, it is imperative that test procedure completeness and adequacy be built into the development process. Therefore, any efforts to tailor this standard must ensure that the procedures generated are both practical to perform and adequate to completely evaluate the equipment or system being tested.

C.2.2 Tailoring to eliminate redundant effort. The development of test outlines and test procedures in accordance with this standard is accomplished in five phases.

a. Definition of the equipment or system to be tested (see 5.2.1).

b. Determination of testing requirements (see 5.2.2 through 5.2.6).

c. Test outline preparation and submittal (see 5.2.7, 5.2.8, and Appendix B).

d. Test procedure development and submittal (see 5.2.9, 5.2.10, and Appendix B).

e. Test procedure verification and final test procedure submittal (see 5.2.11 through 5.2.12.2).

C.2.3 Documentation of equipment and systems after minor modification. In cases where test outlines and test procedures have been prepared and validated for an equipment or system which subsequently undergoes minor modification, the following direction shall be used.

a. Define the differences between the modified and unmodified equipment or system.

b. List any functions, characteristics, or interfaces that have been deleted as a result of the modification and delete the corresponding test elements from the test outlines and test procedures.

c. List any functions, characteristics, or interfaces that have been added as a result of the modification and determine what test elements must be added to reflect these changes by using the instructions provided in 5.2.2 and 5.2.3.

d. List any changes in EXPECTED RESULTS SHEETS of existing test procedures is a result of equipment or system modifications.

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e. Incorporate the changes resulting from b, c, and d above into existing test outlines and test procedures. Document the test outlines and test procedures in the format prescribed by Appendix B and in accordance with 5.2.8 and 5.2.10 of this standard.

f. Verify final test procedures in accordance with 5.2.11 of this standard.

C.2.4 Deviations from the standard. The requirements of this standard may be altered to reduce costs and shorten the development time by permitting deviations that do not alter the basic development processes. Examples of such deviations are:

a. Permitting the use of different formats for documenting the development process. Processes such as interface analysis may be accomplished with the aid of automated data processing and as a result differ from the Interface Analysis format. A variety of formats may be acceptable for documenting the development process; however, no deviation should be permitted to the final test outline and test procedure formats specified in Appendix B.

b. Contracting organization review and approval points may be changed to accommodate difference in program complexity and experience level of TDA.

c. The amount of supporting documentation that is submitted to the contracting organization may be changed by tailoring the Contract Data Requirements List (CDRL).

d. Requirements for type style, reproducible copy, and preparation for delivery may be changed or waived to accommodate individual program requirements.

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